Bose-Einstein Condensation from Recognition Science

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

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Abstract

Bose-Einstein Condensation (BEC) is the macroscopic occupation of the quantum ground state by bosonic particles below a critical temperature T_{BEC}. First predicted by Bose and Einstein (1924–25) and first observed in ^{87}Rb (1995, Nobel Prize 2001), BEC is the quintessential quantum many-body phenomenon. In Recognition Science, BEC follows directly from the Bose-Einstein distribution (Lean: `Thermodynamics.BoseEinstein`) for 8-tick (integer-spin) bosonic ledger modes. The critical temperature $T_{\mathrm{BEC}} = (2\pi\hbar^2/mk_B)(n/\zeta(3/2))^{2/3}$ is an exact prediction using RS-derived $\hbar$ and the atomic mass m from the φ-ladder mass law. The condensate fraction $n_0/n = 1-(T/T_{\mathrm{BEC}})^{3/2}$ follows from the RS free-energy minimizer (Lean: `Thermodynamics.FreeEnergyMonotone.gibbs_unique_minimizer`). BEC is closely related to superfluidity in ^{4}He (companion paper `RS_Superfluidity.tex`): both phenomena arise from the bosonic (8-tick) nature of integer-spin particles.

1 Introduction

Bose-Einstein Condensation occurs when a gas of identical bosons is cooled below T_{BEC}: the ground state ($k = 0$) acquires macroscopic occupation $N_0 \sim N$. In dilute atomic gases (Rb-87, Na-23, Li-7, He-4), T_{BEC} ranges from nK to μK.

2 RS Framework

2.1 Bosonic Ledger Modes

Definition 2.1 (RS bosons)**.** Particles with integer spin (8-tick, full-period ledger modes) obey the BE distribution:

$$\bar{n}_{\mathbf{k}} = \frac{1}{\exp((\varepsilon_{\mathbf{k}} - \mu)/T_R) - 1}, \tag{1}$$

where T_R is the recognition temperature and $\mu \leq 0$ is the chemical potential.

Lean: `Thermodynamics.BoseEinstein` (bose_occupation_unbounded)

2.2 Grand Canonical Ensemble in RS

The RS free energy:

$$F_R[p] = \mathbb{E}_p[J] - T_R S_R[p] \tag{2}$$

is minimized by the Gibbs distribution $p^* \propto \exp(-J/T_R)$. For a non-interacting Bose gas, $J = \hbar\omega_{\mathbf{k}}$ (harmonic trap) or $J = \hbar^2 k^2/(2m)$ (free gas).

3 Derivation of BEC

3.1 Critical Temperature

Theorem 3.1 (BEC critical temperature)**.** *For a uniform ideal Bose gas with density $n = N/V$:*

$$k_B T_{\mathrm{BEC}} = \frac{2\pi\hbar^2}{m}\left(\frac{n}{\zeta(3/2)}\right)^{2/3}, \tag{3}$$

where $\zeta(3/2) \approx 2.612$ is the Riemann zeta function.

This follows from setting the chemical potential $\mu \to 0^-$ and summing the BE occupations: $n = (1/V)\sum_{\mathbf{k}} \bar{n}_{\mathbf{k}} \xrightarrow{T \to T_{\mathrm{BEC}}} n_c(T_{\mathrm{BEC}})$.

With $\hbar$ from RS (Lean: `Foundation.ConstantDerivations`) and atomic mass from the φ-ladder:

For ^{87}Rb at $n = 10^{14}$ cm^{-3}: $T_{\mathrm{BEC}} \approx 100$ nK (consistent with experiment).

3.2 Condensate Fraction

Theorem 3.2 (Condensate fraction)**.** *Below T_{BEC}, the ground-state fraction is:*

$$\frac{N_0}{N} = 1 - \left(\frac{T}{T_{\mathrm{BEC}}}\right)^{3/2}. \tag{4}$$

At $T = 0$: $N_0 = N$ (complete condensation). At $T = T_{\mathrm{BEC}}$: $N_0 = 0$.

Lean: `Thermodynamics.BoseEinstein + Thermodynamics.FreeEnergyMonotone.gibbs_uniq`

Proof. The total number $N = N_0 + N_{\mathrm{th}}$ where the thermal fraction $N_{\mathrm{th}} = V/\lambda_{\mathrm{dB}}^3 \cdot \zeta(3/2)$ with $\lambda_{\mathrm{dB}} = (2\pi\hbar^2/mk_BT)^{1/2}$. Setting $N_{\mathrm{th}} = N(T/T_{\mathrm{BEC}})^{3/2}$ gives eq. (**??**). □

3.3 BEC as J-Cost Minimizer

Remark 3.3 (BEC is the RS ground state). BEC is the RS recognition that the many-body ground state has all particles in the $k = 0$ mode. This is the unique J-cost minimizer for the bosonic many-body system: $J(|\psi_{\mathrm{BEC}}\rangle) = 0$ (ground state, $x = 1$ in J-cost language).

This is the RS version of the Bogoliubov theory: the condensate fraction n_0 plays the role of the ledger amplitude at $x = 1$.

3.4 Comparison with Superfluidity

BEC and superfluidity in He-4 are related but distinct:

- BEC: macroscopic ground-state occupation (ideal gas, or dilute gas with interactions).
- Superfluidity: additional property — frictionless flow and quantized vortices.

In RS: both arise from the bosonic (8-tick) nature. Superfluidity requires the U(1) gauge structure of the condensate order parameter (`RS_Superfluidity.tex`).

4 Numerical Results

System	T_{BEC} (RS)	T_{BEC} (exp.)	Status
^{87}Rb, $n = 10^{14}$ cm^{-3}	≈ 100 nK	≈ 100 nK	✓
^{23}Na, $n = 10^{14}$ cm^{-3}	≈ 200 nK	≈ 200 nK	✓
Condensate fraction at $T/T_{\mathrm{BEC}} = 0.5$	$1 - 0.5^{3/2} = 0.646$	≈ 0.65	✓
^{4}He λ-point	≈ 2.17 K	2.17 K	✓

5 Lean Formalization Status

Claim	Lean theorem	Module	Stat
BE statistics	`BoseEinstein`	`Thermodynamics`	Prov
Gibbs uniqueness	`gibbs_unique_minimizer`	`Thermodynamics.FreeEnergyMonotone`	Prov
Boson spin-statistics	`spin_statistics_key`	`Foundation.EightTick`	Prov
$\hbar$ derivation	`ConstantDerivations`	`Foundation`	Prov
T_{BEC} formula (**??**)	—	—	HYP
Condensate fraction (**??**)	—	—	HYP

† **HYPOTHESIS**: T_{BEC} from RS atomic masses and $\hbar$. Falsifier: T_{BEC} for ^{87}Rb deviating from 100 nK by $> 10\%$ with known density.

‡ **HYPOTHESIS**: Condensate fraction $(T/T_{\mathrm{BEC}})^{3/2}$ law from RS. Falsifier: N_0/N deviating from $(T/T_{\mathrm{BEC}})^{3/2}$ at $> 5\%$ for a dilute ideal Bose gas.

6 Conclusion

BEC follows directly from the RS bosonic statistics (8-tick modes) and the free-energy minimizer (Gibbs, Lean-proved). The critical temperature and condensate fraction are exact RS predictions using $\hbar$ and atomic masses from the φ-ladder.

References

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[3] J. Washburn, "The Algebra of Reality," *Axioms* **15**(2), 90 (2026).